

BAM Reference Data: Creep of Single-Crystal Ni-Based Superalloy CMSX-6 Version 1.0 (October 2024)

This publication provides comprehensive metadata and test results of constant force creep tests according to DIN EN ISO 204:2019-4 on the single crystal Ni-based superalloy CMSX-6 at $T = 980\text{ °C}$ and initial stresses between 140 MPa and 230 MPa. The tests were carried out in an ISO 17025 accredited test laboratory. The calibrations of measuring equipment are documented, meet the requirements of the measurement standard and are metrologically traceable. The provided data were audited and are BAM reference data.

The data can be useful for checking own creep test results on nominally similar material, for verifying the own testing set-up (e.g. by testing same/similar material) and as input data for simulations of creep behavior for design and for alloy development. There are currently no known restrictions on the use of the data.“

The data are provided using v1.1 of the “NFDI-MatWerk/IUC02 Data schema for creep data of Ni-based superalloys including a comprehensive documentation of test results and metadata” [1] as a template for data documentation and structuring. The data schema has an overarching structure (Columns B to E) which mimics the way a domain expert would structure the data (see **Figure 1**) [1]. The provided data fully meets the requirement profile (mandatory fields in column K of the data schema). The requirement profile was agreed between domain experts.

The provided XLSX file “2024-10_Creep-reference-data-CMSX-6_v1.0.xlsx” contains the data schema (data sheet “Data-Schema-Creep_v1.1”) and additional worksheets named by the respective test ID, and which provide in column G the answers to the individual entries (column C) of the schema. Some answers reference to further worksheets in the same XLSX file containing the respective metadata or test results. Additionally, we provide LIS Files (ASCII format), which contain a header with selected mandatory metadata and test results, and the mandatory data series. Furthermore, we provide selected PDF files regarding the as-tested material and the test piece. Overall, the decimal point setting is based on the specifications of DIN EN ISO 204:2019-4 and the resolution of the measuring devices. The measurement uncertainty of test temperature (T), initial stress (R_o) and percentage plastic extension from end of loading (e) were determined according to “Measurement uncertainties in mechanical tests on metallic materials- Part 3 - CWA 15261-3:2005” and are shown in **Table 1**.

Table 1. Combined standard uncertainties (confidence level 68 %) for test temperature (T), initial stress (R_o) and percentage plastic extension from end of loading (e).

$\pm u_c\%$ (T) [$^{\circ}\text{C}$]	$\pm u_c\%$ (R_o) [MPa]	$\pm u\%$ (e) [%]
%	%	%
Max. 0.30	0.67	1.0

Metadata	Test info	Test job details	
		Test parameters	
		Related research outcome	
	Material related	History and condition of the material	As-manufactured material
			As-tested material
			Heat treatment
			Chemical composition
			Microstructure
			Results from NDT
			Results from other mechanical tests
			...
		Test piece	
	Measuring and test equipment	Test machine	Data acquisition (e.g., unit)
			...
		Test force	Loading system
			Load sensor
			Data acquisition (e.g., frequency)
		Laboratory conditions	
		Temperature-measuring system	Measuring instrument
			Data acquisition (e.g., calibrated range)
			...
		Extensometer system	Contacting extensometer
			Non-contacting extensometer
		...	
		Data processing procedures	
	Primary data	Test result	Values recorded at test start
Values recorded during test run			
Values recorded after end of test			
Secondary data	Test result	Values recorded during test run	
		Elongation values	
		Extension values	

Figure 1. Data schema structure. The shown excerpt of the full data schema illustrates the four overarching levels of categorization, corresponding to columns B to E of data sheet “Data-Schema-Creep_v1.1” in the attached XLSX file. Reproduced from Ávila Calderón et al. [1] under license CC BY 4.0 (Zenodo, 2024).

References

[1] L. A. Ávila Calderón, Y. Shakeel, S. Schriever, A. Gedsun, M. Forti, H. Jordan, Y. Han, O. Baer, R. Aversa, J. Olbricht, T. Hammerschmidt, T. Hickel, & B. Skrotzki, NFDI-MatWerk/IUC02 Data schema for creep data of Ni-based superalloys including a comprehensive documentation of test results and metadata (1.1), Zenodo (2024), DOI: 10.5281/zenodo.11668375

Acknowledgment

This work was carried out in the context of NFDI-MatWerk and partially funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under the National Research Data Infrastructure – NFDI 38/1 – project number 460247524. The authors acknowledge Mariano Forti from Ruhr Universität Bochum for fruitful discussions on data preparation for further usage, and Kathrin Matzak from BAM for providing the $R_{p0.2}$ values.